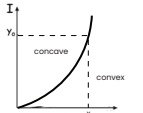


CURRENT ELECTRICITY

01 ELECTRIC CURRENT

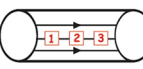
Current $I = \frac{q}{t}$
 Average current $I_{av} = \frac{\Delta q}{\Delta t}$
 Instantaneous current $I_{inst} = \frac{dq}{dt}$
 Average current I_{av}
 $= \frac{\text{area under I-t graph}}{\text{total time taken}}$

 Concave area $= \frac{2}{3} \times y_0 y_0$
 Convex area $= \frac{1}{3} \times y_0 y_0$

02 DRIFT VELOCITY

Free Electron

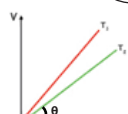
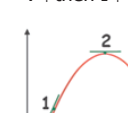
 $\langle KE \rangle = \frac{3}{2} kT$
 $< \frac{1}{2} mv^2 > \approx 10^{-21} J$
 Avg. Speed = 105 m/s
 Electrons are in random motion
 $\vec{v}_1 + \vec{v}_2 + \vec{v}_3 + \dots + \vec{v}_n$
 $\vec{v}_{net} = 0$
 $\vec{v} = \vec{u} + a\tau$
 $v_d = a\tau$
 $v_d = \frac{eE}{m} \tau$
 $E = \frac{V}{l}$
 $v_d = \frac{eV}{mI} \tau$

03 FACTORS AFFECTING DRIFT VELOCITY

- Dependence on shape
 1) Uniform shape

 $v_d = \frac{eE}{m} \tau$
 Here E is uniform so,
 $v_{d1} = v_{d2} = v_{d3}$ $E_1 = E_2 = E_3$
 2) Non Uniform shape

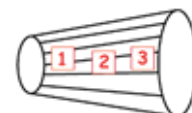
 $E_1 > E_2 > E_3$
 $v_d \propto E$ $v_{d1} > v_{d2} > v_{d3}$
 3) Relation B/w Current & Drift velocity
 $I = nA v_d e$
 $n = \text{no. of e's per unit volume}$

04 OHM'S LAW

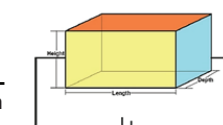
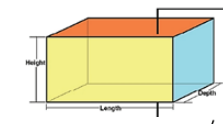
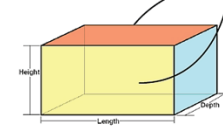
$V = I \times R$
 Voltage (V) Current (A) Resistance (ohms)

 $\text{Slope} = \tan \theta = R$
 $R = \frac{\rho l}{A}$
 $\rho = \frac{RA}{l} = \frac{m}{ne^2 \tau}$
 Depends on
 1. Material (n & τ changes)
 2. Temperature (n & τ changes)
 3. Dimension (Length & Area)
 Non-Ohmic Conductor
 V-I graph is not linear
 Slope of tangent $\frac{dv}{dI} = R$
 Resistance is not constant
 1) Slope = +ve Resistance = +ve
 $V \uparrow \text{ then } I \uparrow$

 2) Slope = 0 Resistance = 0
 3) Slope = -ve Resistance = -ve
 $V \uparrow \text{ then } I \downarrow$

05 CURRENT DENSITY

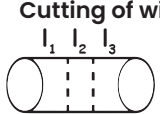
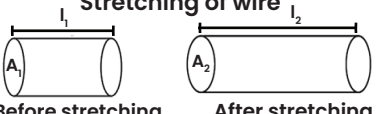
$J = \frac{I}{A}$

 $I = \frac{V}{R} = \frac{V}{\rho l / A} = \frac{EA}{\rho}$
 $J = \frac{E}{\rho}$, $J \propto E$
 Uniform cross section
 $E_1 = E_2 = E_3$
 $\therefore J_1 = J_2 = J_3$
 Non-Uniform cross-section
 $E_1 > E_2 > E_3$ $J \propto \frac{1}{A}$
 $\therefore J_1 > J_2 > J_3$
 But current is same
 $I_1 = I_2 = I_3$


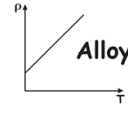
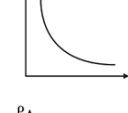
06 DEPENDANCE OF R ON DIMENSION

$R = \frac{\rho l}{A} = \frac{\rho l}{bh}$

 $R = \frac{\rho h}{lb}$

 $R = \frac{\rho b}{lh}$

 $\frac{R_{max}}{R_{min}} = \frac{(\text{max length})^2}{(\text{min length})^2}$

07 CUTTING & STRETCHING OF WIRE

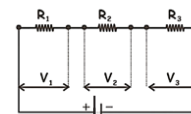
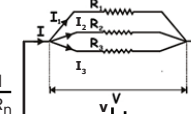
Cutting of wire
 $R_1 \propto l_1$
 $R_2 \propto l_2$
 $R_3 \propto l_3$

 Stretching of wire

 Before stretching After stretching
 $\frac{R_1}{R_2} = \left(\frac{l_1}{l_2} \right)^2 = \left(\frac{A_2}{A_1} \right)^2 = \left(\frac{r_2}{r_1} \right)^4 = \left(\frac{d_2}{d_1} \right)^4$
 If l become nl then R become $n^2 R$
 If r become r/n then R become $n^4 R$
 If change in length $> 10\%$
 $\frac{R_2 - R_1}{R_1} \times 100 = \frac{l_2^2 - l_1^2}{l_1^2} \times 100$
 If change in length $< 10\%$
 1) % change in $R = 2 \times$ % change in length
 2) % change in $R = 2 \times$ % change in area
 3) % change in $R = 4 \times$ % change in radius

08 TEMPERATURE DEPENDANCE OF RESISTANCE

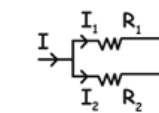
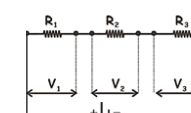
Metals, $\alpha = +ve$
 If $T \uparrow, R \uparrow$

 Alloy

 For semiconductor $\alpha = -ve$
 If $T \uparrow, R \downarrow$

 $\alpha = \text{slightly increasing with temp.}$
 Resistance slightly increases with T
 Variation of resistance with temperature
 $\alpha = \frac{\Delta R}{R \Delta T}$
 Equivalent temp. Coefficient
 SERIES:
 $\alpha_s = \frac{\alpha_1 R_1 + \alpha_2 R_2}{R_1 + R_2}$
 if $R_1 = R_2$ $\alpha = \frac{\alpha_1 + \alpha_2}{2}$
 PARALLEL:
 $\alpha_p = \frac{\frac{\alpha_1}{R_1} + \frac{\alpha_2}{R_2}}{\frac{1}{R_1} + \frac{1}{R_2}}$
 if $R_1 = R_2$ $\alpha = \frac{\alpha_1 + \alpha_2}{2}$

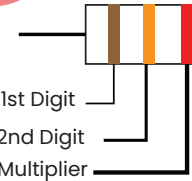
09 GROUPING OF RESISTANCE

Series Combination
 Current is constant
 voltage is divided
 $R_s = R_1 + R_2 + R_3 + \dots + R_n$

 If resistors are identical: $R_s = nR$
 Parallel Combination
 voltage is constant
 current is divided
 $\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots + \frac{1}{R_n}$

 If resistors are identical: $R_p = \frac{R}{n}$
 Shortcut for two resistors in parallel
 $\frac{R_1 R_2}{R_1 + R_2}$
 R_1 } R_s Bigger than largest value of resistance
 R_2 } R_p Lower than smallest value of resistance

10 CURRENT & VOLTAGE DIVIDER RULE

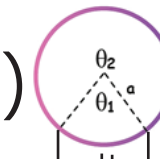
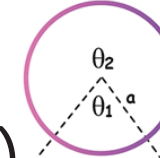
Current Divider Rule
 $V = \text{Constant}$ $I \propto \frac{1}{R}$

 $I_1 = \frac{I \times R_2}{R_1 + R_2}$, $I_2 = \frac{I \times R_1}{R_1 + R_2}$
 Voltage Divider Rule
 $V_1 = IR_1$ $V_2 = IR_2$ $V_3 = IR_3$
 $V_1 = \frac{V \times R_1}{R_1 + R_2 + R_3}$
 $V_2 = \frac{V \times R_2}{R_1 + R_2 + R_3}$
 $V_3 = \frac{V \times R_3}{R_1 + R_2 + R_3}$


11 COLOUR CODING

1st Digit
 2nd Digit
 Multiplier
 Tolerance

 Resistor color code

Color	Digit	Multiplier	Tolerance (%)
Black	0	10^0	
Brown	1	10^1	1
Red	2	10^2	2
Orange	3	10^3	
Yellow	4	10^4	
Green	5	10^5	0.5
Blue	6	10^6	0.25
Violet	7	10^7	0.1
Grey	8	10^8	
White	9	10^9	
Gold		10^{-1}	5
Silver		10^{-2}	10
None			20

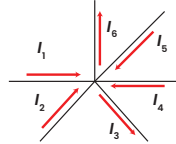
12 GEOMETRICAL DIAGRAM

Circle formed by wire having uniform resistance per unit length (r)
 $R_{eff} = r\alpha \left(\frac{\theta_1 \theta_2}{\theta_1 + \theta_2} \right)$

 When resistance of wire forming circle is given
 $R_{eff} = \frac{R}{2\pi} \left(\frac{\theta_1 \theta_2}{\theta_1 + \theta_2} \right)$


13

KIRCHHOFF'S LAW

1. Junction Rule



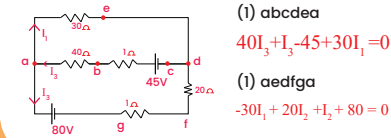
$$\sum I_{in} = \sum I_{out}$$

$$I_1 + I_2 + I_4 + I_5 = I_3 + I_6$$

2. Open Circuit

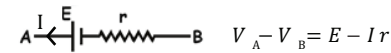
- (1) A $\rightarrow$ B $V_A - V_B = -E$
 (2) A $\rightarrow$ B $V_A - V_B = +E$
 (3) A $\rightarrow$ B $V_A - V_B = IR$
 (4) A $\rightarrow$ B $V_A - V_B = -IR$

Closed Circuit



14

CELL & INTERNAL RESISTANCE



Terminal potential difference (TPD)

1) When current is drawn from cell

$$V = V_A - V_B = E - Ir$$

$$TPD < EMF$$

$$V = E - Ir = IR \Rightarrow I = \frac{E}{R + r}$$

$$\text{So, } V = \frac{E}{R + r} R$$

2) When current is given to cell

$$V = V_A - V_B = E + Ir$$

$$TPD > EMF$$

15

CELL & INTERNAL RESISTANCE

3) When cell is in open circuit

$$A \rightarrow B \quad \text{Here, } I = 0$$

$$TPD = E$$

3) When cell is in short circuit

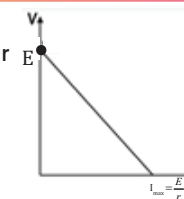
$$I = I_{max} = \frac{E}{r}$$

$$TPD = 0$$

16

CELL & INTERNAL RESISTANCE

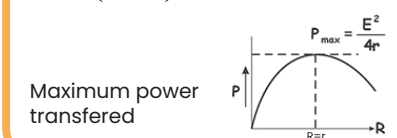
Slope of graph = $-r$
 y intercept = E



$$\text{Internal Resistance } r = \left(\frac{E - V}{V} \right) R$$

Power delivered by cell during withdrawal of current

$$P = I^2 R = \left(\frac{E}{R + r} \right)^2 R \quad P_{max} = \frac{E^2}{4r} \text{, when } R = r$$

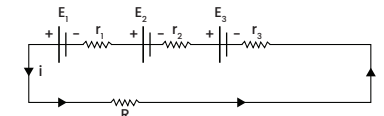


Maximum power transferred

17

COMBINATION OF CELLS

1) Series Combination



(a) $E_{equivalent} = E_1 + E_2 + E_3 + \dots + E_n$

(b) $r_{equivalent} = r_1 + r_2 + r_3 + \dots + r_n$

(c) Current, $i = \frac{\sum E_i}{\sum r_i + R}$

(d) If all cells have equal emf E and equal internal resistance r then $i = \frac{nE}{nr + R}$

1) If $nr \gg R \Rightarrow i = \frac{E}{r}$

2) If $nr \ll R \Rightarrow i = \frac{nE}{R}$

(e) Power dissipated in circuit $P = i^2 R = \left(\frac{nE}{nr + R} \right)^2 R$

Conditions for maximum power: $R = nr$

$P_{max} = nE^2/4r$

19

COMBINATION OF CELLS

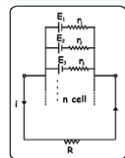
18

COMBINATION OF CELLS

2) Parallel Combination

a) $E_{equivalent} = \frac{E_1 + E_2 + E_3 + \dots + E_n}{\frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3} + \dots + \frac{1}{r_n}}$

b) $r_{equivalent} = \frac{1}{\frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3} + \dots + \frac{1}{r_n}}$

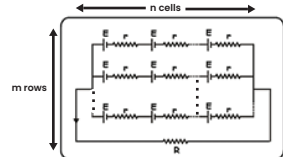


(c) If all cells have equal emf E & internal resistance r then $E_{equivalent} = E$

$$r_{equivalent} = \frac{r}{n} \Rightarrow \text{current } i = \frac{E}{\frac{r}{n} + R}$$

(d) Power dissipated in the circuit $P = I^2 R = \left(\frac{nE}{r + nR} \right)^2 R$

3) Mixed Combination



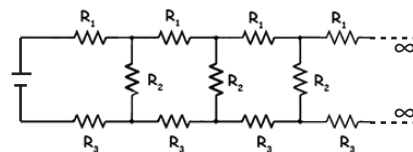
$$r_{net} = \frac{nr}{m} \quad \text{Total emf} = E_{net} = nE$$

$$I = \frac{E_{net}}{R + r_{net}} = \frac{nE}{R + \frac{nr}{m}}$$

$$P_{max} = \frac{E_{eq}^2}{4R} = \frac{mnE^2}{4r} \quad [\because R = \frac{nr}{m}]$$

n cells connected in series and there are m such branches in the circuit.
 Internal resistance of cells connected in a row = nr

Infinite resistors



$$R_{eq} = \frac{R_1 + R_3}{2} \left[1 + \sqrt{1 + \frac{4R_2}{R_1 + R_3}} \right]$$

If all resistors are equal $R_{eq} = R(1 + \sqrt{3})$

20

3D CIRCUIT

Edge $R_{eq} = \frac{7R}{12}$

Face $R_{eq} = \frac{3R}{4}$

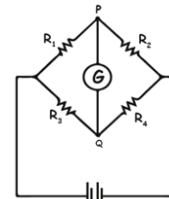
Body $R_{eq} = \frac{5R}{6}$

21

WHEATSTONE BRIDGE

1) Balanced WSB

Balanced Condition $\frac{R_1}{R_2} = \frac{R_3}{R_4}$
 $V_P = V_Q$
 Current through $G = 0$

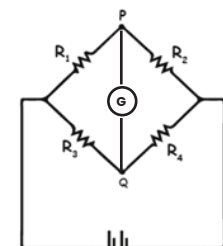


2) Unbalanced WSB

$V_P \neq V_Q$

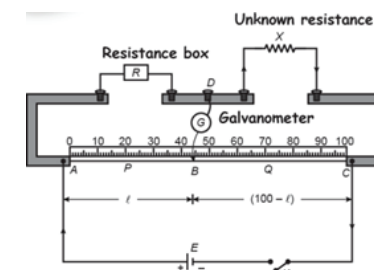
if $\frac{R_1}{R_2} > \frac{R_3}{R_4}$

then $V_Q > V_P$



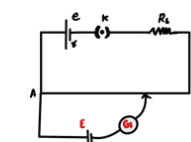
22

METER BRIDGE



$$\frac{P}{Q} = \frac{R}{X} = \frac{\ell}{100 - \ell}$$

POTENTIOMETER



23

POTENTIOMETER

POTENTIAL

$$V_{AB} = \left(\frac{e}{r + R_h + R} \right) R$$

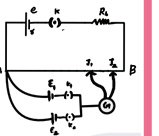
where, R = Resistance of potentiometer wire

POTENTIAL GRADIENT

$$x = \frac{V_{AB}}{L} = \frac{eR}{r + R_h + R} = \left(\frac{e}{r + R_h + R} \right) \frac{R}{L}$$

1. COMPARISON OF CELL

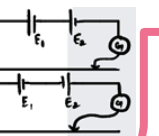
$$\frac{E_1}{E_2} = \frac{l_1}{l_2}$$



2. BOTH BATTERIES ARE CONNECTED TOGETHER

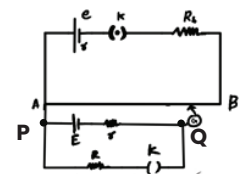
(Once with same polarity then with opposite polarity)

$$\frac{E_1}{E_2} = \frac{l_1 + l_2}{l_1 - l_2}$$



24 POTENTIOMETER

3. CALCULATION OF INTERNAL RESISTANCE



$$r = R \left(\frac{E}{V} - 1 \right) = R \left(\frac{l_1}{l_2} - 1 \right)$$

$E = V_{PQ}$ when key is open

$V = V_{PQ}$ when key is close

25 HEATING EFFECT OF ELECTRIC CURRENT

POWER

$$P = \frac{dH}{dt} = VI = \frac{V^2}{R} = I^2R$$

ELECTRIC KETTLE

Time taken for first coil- t_1 , time taken for second coil- t_2

if they are connected in series $t_s = t_1 + t_2$	if they are connected in parallel $t_p = \frac{t_1 t_2}{t_1 + t_2}$
--	--

BULB

$$P_{\text{rated}} = \frac{V_{\text{rated}}^2}{R} \Rightarrow R = \frac{V_{\text{rated}}^2}{P_{\text{rated}}}$$

CONNECTED IN SERIES



$$P_{\text{dissipated}} = I^2 R$$

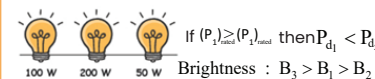
$$P_{\text{dissipated}} \propto R$$

$$\text{Brightness} \propto R$$

26 HEATING EFFECT OF ELECTRIC CURRENT

$$P_{\text{dissipated}} \propto \frac{(V_{\text{rated}})^2}{P_{\text{rated}}}$$

$$P_d (\text{brightness}) \propto \frac{1}{P_{\text{rated}}}$$



CONNECTED IN PARALLEL

$$P_{\text{rated}} = \frac{(V_{\text{rated}})^2}{R}$$

$$P_d = \frac{V^2}{R} \Rightarrow P_d \propto \frac{1}{R}$$

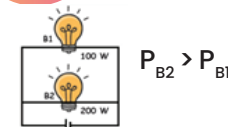
$$\Rightarrow P_d (\text{brightness}) \propto \frac{1}{(V_{\text{rated}})^2} \Rightarrow P_d (\text{brightness}) \propto \frac{P_{\text{rated}}}{(V_{\text{rated}})^2}$$

$$P_d (\text{brightness}) \propto P_{\text{rated}}$$

$$\text{if } (P_1)_R > (P_2)_R$$

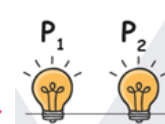
$$\Rightarrow \text{Brightness } (P_d)_1 > (P_d)_2$$

27 HEATING EFFECT OF ELECTRIC CURRENT



COMBINATION OF BULBS

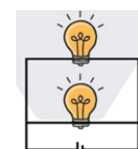
SERIES



TOTAL POWER

$$P = \frac{P_1 P_2}{P_1 + P_2}$$

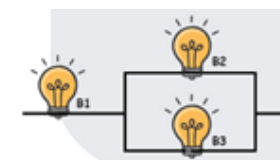
PARALLEL



$$P = P_1 + P_2$$

28 HEATING EFFECT OF ELECTRIC CURRENT

FUSED BULB



If bulb 2 is fused then,

for bulb 3

$R \uparrow$

$V \uparrow$

$P \uparrow$

$B \uparrow$

for bulb 1

$V \downarrow$

$P \downarrow$

$B \downarrow$

If bulb is added in parallel:-

for example if bulb 3 is added in parallel to 2 then

for bulb 3

$R \downarrow$

$V \downarrow$

$P \downarrow$

$B \downarrow$

for bulb 1

$V \uparrow$

$P \uparrow$

$B \uparrow$

29 CONVERSION OF GALVANOMETER

Current sensitivity

$$S_i = \frac{\theta}{I}$$

where, θ = angle of deflection in galvanometer
 I = Corresponding current in galvanometer

Unit: $\frac{\text{divisions}}{\text{ampere}}$ OR $\frac{\text{rad}}{\text{A}}$

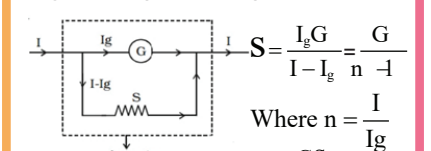
Voltage sensitivity

$$S_v = \frac{\theta}{V}$$

where, θ = angle of deflection in galvanometer
 V = Corresponding voltage across galvanometer

$\frac{\text{divisions}}{\text{voltage}}$ OR $\frac{\text{rad}}{\text{V}}$

GALVANOMETER TO AMMETER



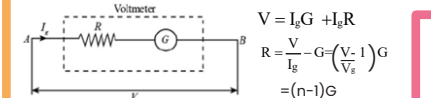
$$S = \frac{I_g G}{I - I_g} = \frac{G}{n - 1}$$

$$\text{Where } n = \frac{I}{I_g}$$

$$\text{RESISTANCE OF AMMETER } R = \frac{GS}{G + S}$$

RESISTANCE OF IDEAL AMMETER = 0

GALVANOMETER TO VOLTmeter



$$R = \frac{V}{I_g} - G = \left(\frac{V}{I_g} - 1 \right) G$$

where, $n = \frac{V}{I_g G}$

RESISTANCE OF VOLTmeter = $G + R$

RESISTANCE OF IDEAL VOLTmeter = ∞